

MSE 360: Kinetic Processes in Materials

Homework #6

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. Equation 4.40 in the text provides the solution for the transient diffusion of a “thin layer” of material between two semi-infinite bodies:

$$C_i(x, t) = \frac{N_i}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)}$$

This thin-film solution yields a Gaussian profile that gradually broadens as a function of time. Derive a mathematical expression quantifying how this Gaussian profile “broadens” as a function of time. Use the peak width at half-maximum (i.e., the distance between the two points where the concentration is one-half of its peak value at any given time) as the definition of peak breadth.